

Assignment 2 – Case Study Lab

Stage 2 Lab Activities

SmartCare Requirements Engineering

AI OFF -> AI ON -> VERIFY | 1 hour

Learning objectives

- Analyse the SmartCare client brief.
- Identify stakeholders and scope.
- Write functional and non-functional requirements.
- Develop user stories and Given-When-Then acceptance criteria.
- Use AI to critique requirements without allowing it to invent stakeholder needs.
- Produce SmartCare Requirements Specification v1.0.

Part A - Client Brief: AI OFF

SmartCare uses spreadsheets and paper records. Staff report duplicate bookings, difficulty finding patient information, inconsistent appointment status and limited appointment history. Management wants a small, maintainable patient, practitioner and appointment system.

SmartCare currently has several operational problems due to workplace practices where records are made both digitally and manually, primarily causing:

- duplicate appointment bookings and patient records
- difficulties locating patient records
- inconsistent appointment statuses, further caused by manual cancellation processes
- difficulties finding practitioner availability
- an unreliable appointment history
- difficulties producing operational reports

As a result, the client requests a simple software system that can improve patient, practitioner and appointment management.

2. Stakeholders

Part B - Stakeholders and Scope: AI OFF

Identify at least four stakeholders. Create In Scope and Out of Scope lists. Label uncertain features as provisional rather than confirmed.

Stakeholders:

- Patient
- Healthcare Practitioner
- Receptionist
- Clinic Manager
- System Administrator

Scope:

In Scope	Out of Scope	Provisional
• Patient registration • Appointment management (creation / cancellations, statuses) • Cancelled appointments remain in history • Prevent of double-booking practitioners • Prevention of duplicate patient records • Patient search • Produce operational reports • Save data during sessions	• Facial recognition login • Online payment • AI-recommended treatments • AI diagnosis recommendations • Insurance processing	• Patient SMS reminders • Appointment management permissions • Practitioner schedule view

Part C - Functional Requirements: AI OFF

Write 8-12 numbered functional requirements using FR-01, FR-02 and so on. Each should describe one observable capability.

FR-01: The system shall allow staff to register patients

FR-02: The system shall allow staff to register practitioners

FR-03: The system shall allow practitioners to view patient information and appointment details

FR-04: The system shall allow receptionists to view practitioner availability

FR-05: The system shall allow receptionists to create appointments

FR-06: The system shall allow receptionists to cancel appointments

FR-07: The system shall prevent staff from allocating appointments at the same time for one practitioner

FR-08: The system shall prevent staff from inputting invalid patient/appointment information

FR-09: The system must maintain relevant patient, appointment and staff data during sessions

FR-10: The system must allow staff to search for patients in the system

FR-11: The system must generate operational reports

FR-12: The system must maintain appointment status information for every patient

Part D - Non-Functional Requirements: AI OFF

Write 4-6 numbered non-functional requirements covering appropriate qualities such as reliability, maintainability, usability, data integrity or testability.

NFR-01: The system must ensure recorded data (patient information, appointment details, etc.) remains in the system even if the system has shut down

NFR-02: The system must handle invalid inputs or unexpected errors whilst ensuring said issues never causes the system to crash

NFR-03: The system must be designed in separated components to allow the possibility of new features to be developed in the future without breaking existing functions.

NFR-04: The system must provide clear and easy-to-understand messages when an input is successful or when an error occurs.

NFR-05: The system must ensure that any updates to data such as appointment statuses are instantly updated across all schedules to prevent inconsistent information

NFR-06: The system must allow individual functions to be tested separately without running the whole system

Part E - User Stories and Acceptance Criteria: AI OFF

Write 4-6 user stories. For at least three, create Given-When-Then acceptance criteria including one negative or failure scenario.

User Stories:

US-01: As a receptionist, I want to register a patient into the system, so that patient information is accessible and recorded for future visits.

US-02: As a practitioner, I want to set my weekly availability time, so that I am not allocated two appointments at the same time for two different patients.

US-03: As a receptionist, I want to search for a patient on the system, so that I can efficiently find the relevant patient information.

US-04: As a receptionist, I want to create appointments, so that patients are allocated to a practitioner.

US-05: As a clinic manager, I want to view clinic operations and data, so that I can monitor clinic performance and appointment operations.

US-06: As a receptionist, I want to cancel appointments, so that practitioner availability is freed up for other patients while keeping appointment information/statuses accurate.

Acceptance Criteria:

Patient Registration Acceptance Criteria

GIVEN: the receptionist is registering a patient into the system

WHEN: the receptionist enters the patient's details (e.g. first name, last name, age, etc.)

THEN: the system adds the information into the list of registered patients and allows staff to view patient details prior to future visits

Patient Search Acceptance Criteria

GIVEN: the receptionist is searching for an existing patient in the system

WHEN: the receptionist submits a search query matching a patient's name

THEN: the system returns all patient records that matches the query string

Negative Acceptance Criteria

GIVEN: the practitioner already has an appointment with a patient at a specified time

WHEN: staff registers another patient with the occupied practitioner at the same specified time

THEN: the system rejects the input and prints an error statement and warning staff that the practitioner is not available

Part F - AI Requirements Review: AI ON

Prompt: Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Copilot reviewed the functional and non-functional requirements that were created in these activities to determine ambiguities, inconsistencies, missing clarification questions and testability. With this in mind, AI identified the following:

Ambiguities:

AI found multiple ambiguities in the requirements, identifying terms that are unclear, subjective, or undefined. For instance, a major ambiguity found in the non-functional requirements is that NFR-04's "clear and easy-to-understand messages" is subjective. For ambiguities in functional requirements, the use of the term "invalid" fails to define what is considered as an invalid condition.

Missing clarification questions:

Following ambiguities, it would have been proficient to follow up the following ambiguous requirements with clarification questions, which are essentially applied to all the functional requirements (FR-01 to FR-012), and non-functional requirements: NFR-01, NFR-03 and NFR-06.

Inconsistencies:

AI identified one inconsistency between FR-09 and NFR-01:

- FR-09: The system must maintain relevant patient, appointment and staff data during sessions
- NFR-01: The system must ensure recorded data (patient information, appointment details, etc.) remains in the system even if the system has shut down

AI stated: "It is unclear whether FR-09 refers to temporary session data management or long-term data persistence. This may cause confusion about which requirement is responsible for data storage behaviour."

Testability:

Due to the unaddressed ambiguities in the requirements, some functionalities can not be tested until said ambiguities are defined, for instance: FR-08: Cannot be fully tested until "invalid" information is defined.

In addition, AI states business rules need to be known to allow consistent implementation and acceptance testing.

Part G - VERIFY the AI Review

Classify each significant AI suggestion as Accepted, Modified, Rejected, or Unverified. Explain the evidence used.

AI suggestion	Evidence?	Decision	Reason	Verification
Confirm that the system must prevent duplicate appointment bookings for a practitioner.	FR-07 Case study: "Duplicate appointment bookings"	Accepted	FR-07 addresses a current issue in the case study regarding "duplicate appointment bookings."	Verified Supported via case study and FR-07
Clarify practitioner access to patient information	FR-03	Modified	FR-03 does not define what patient information includes	Partially verified Confirms patient information is needed, but does not specify the required details. Therefore, clarification by the client is required
Define appointment status values	FR-12 Case study: "Inconsistent appointment status information"	Modified	FR-12 does not specify appointment status values	Partially verified Confirms appointment status details are required by the client, but status types are not defined. Therefore, clarification by the client is required
Confirm the handling of	FR-06	Unverified	Case study does not specify how	Not verified

cancelled appointments			cancelled appointments are handled	Neither case study or requirements specify how cancelled appointments are to be addressed. Therefore, confirmation by the client is required
Confirm whether operational reports should be defined more precisely before implementation	FR-11 Case study: "Difficulty producing basic operational reports"	Modified	FR-11 states the system must generate "operational reports", however the report contents are not specified	Partially verified Confirms operational reports are required via client needs, but details regarding the report contents are not specified. Therefore clarification by the client is required.

Reflection

In 150-250 words: What did AI notice that you missed? What did AI invent or overreach on? Which requirement changed after review? Why must requirements have evidence?

Despite previous activities, what AI noticed that I missed were multiple ambiguities in the functional and non-functional requirements, in which multiple vague and undefined terms were used to describe requirements. Due to this issue, many confirmations/clarifications need to be made with the client to further ensure said ambiguities are addressed.

During the AI review, AI did not seem to invent any new requirements or overreach.

A requirement that changed after AI's review was FR-09 and NFR-01's wording. An inconsistency was found in the requirements between FR-09 and NFR-01 that failed to define whether data was kept permanently or during sessions. The following are the new refined requirements:

FR-09: The system must save relevant patient, appointment and staff data entered into the system

NFR-01: The system must ensure recorded data (patient information, appointment details, etc.) remains in the system even if the system has been shut down, restarted or if an invalid operation occurs.

Requirements need evidence to ensure that features that are to be implemented prior to creating the project to align with the existing needs requested by the client. By using evidence to prove requirements, it ultimately prevents scope creeps, resource wastage (e.g. money and time), and improves testing to easily determine whether the requirement has been met.